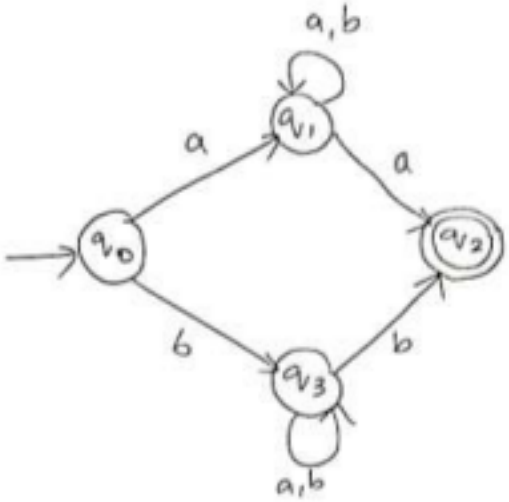
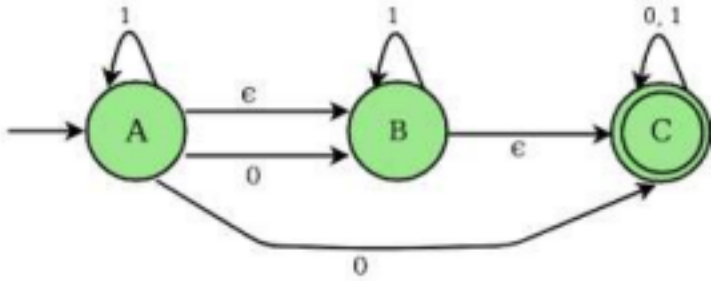
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING

CSA1311 – THEORY OF COMPUTATION

ASSIGNMENT – 1

		CO	BL
1	Design a Deterministic Finite Automata (DFA) to accept strings that start and end with same symbols over □□={0,1}. Write the formal definition of the DFA. Also show that the string **010100** is accepted by the DFA	1	3
2	Design NFA to accept string having 101 as substring	1	3
3	Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true. $$\sum_{□□=1}^{□□} □□(□□+1) = \frac{□□(□□+1)(□□+2)}{3}$$	1	1
4	Consider the NFA given below which accepts the language representing the set of all strings that start and end with the same symbol over the alphabet □□={a,b}. Construct a DFA that represents the same language 	1	3
5	A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states	1	3



1. *Design a DFA to accept strings that start and end with the same symbol over Σ = {0,1}. Write the formal definition of the DFA. Also show that the string 010100 is accepted by the DFA.*

Idea

The DFA must remember the first symbol read and the last symbol read. Each state records both pieces of information:

- q_0 – start state, nothing read yet.
- q_1 – first symbol was 0 and the last symbol read is 0 (accepting).
- q_2 – first symbol was 0 and the last symbol read is 1 (non-accepting).
- q_3 – first symbol was 1 and the last symbol read is 1 (accepting).
- q_4 – first symbol was 1 and the last symbol read is 0 (non-accepting).

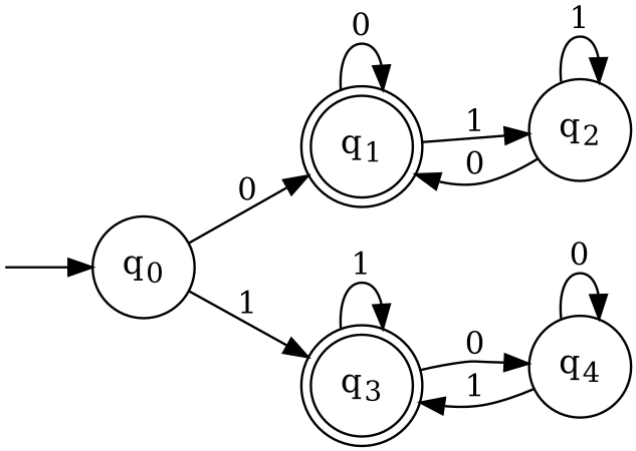
Formal definition

M = (Q, Σ, δ, q_0, F), where

- Q = {q_0, q_1, q_2, q_3, q_4}
- Σ = {0, 1}
- q_0 is the start state
- F = {q_1, q_3}
- δ is given by the transition table below.

δ	**0**	**1**
→ q_0	q_1	q_3
* q_1	q_1	q_2
q_2	q_1	q_2
* q_3	q_4	q_3
q_4	q_4	q_3

Transition diagram



Acceptance of 010100

$\delta(q_0, 0) = q_1$

$\delta(q_1, 1) = q_2$

$\delta(q_2, 0) = q_1$

$\delta(q_1, 1) = q_2$

$\delta(q_2, 0) = q_1$

$\delta(q_1, 0) = q_1$

Path: $q_0 \rightarrow(0)\ q_1 \rightarrow(1)\ q_2 \rightarrow(0)\ q_1 \rightarrow(1)\ q_2 \rightarrow(0)\ q_1 \rightarrow(0)\ q_1$. After reading the whole string the DFA is in $q_1 \in F$, so 010100 is accepted. (It starts with 0 and ends with 0.)

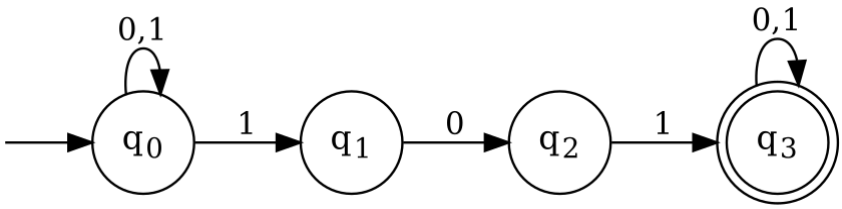
2.Design NFA to accept string having 101 as substring.

Idea: stay in the start state reading anything, guess non-deterministically where 101 begins, then verify 1, 0, 1 in order. Once 101 has been seen, stay in the final state.

N = (Q, Σ, δ, q_0, F), where Q = {q_0, q_1, q_2, q_3}, Σ = {0, 1}, start state q_0, F = {q_3}.

δ	**0**	**1**
→ q_0	{q_0}	{q_0, q_1}
q_1	{q_2}	∅
q_2	∅	{q_3}
* q_3	{q_3}	{q_3}

Transition diagram



Example: for 01101 the accepting path is $q_0 \rightarrow(0)\ q_0 \rightarrow(1)\ q_0 \rightarrow(1)\ q_1 \rightarrow(0)\ q_2 \rightarrow(1)\ q_3$, so the string is accepted. A string such as 1100 has no path to q_3 and is rejected.

3.Prove by mathematical induction that for all positive integers n: ∑ (k = 1 to n) k(k + 1) = n(n + 1)(n + 2) / 3.

Let P(n) be the statement: $1\cdot2 + 2\cdot3 + \ldots + n(n + 1) = n(n + 1)(n + 2) / 3$.

Basis step (n = 1)

LHS = 1(1 + 1) = 2. RHS = 1·2·3 / 3 = 2. LHS = RHS, so P(1) is true.

Inductive hypothesis

Assume P(m) is true for some positive integer m: $\sum (k = 1 \text{ to } m)\ k(k + 1) = m(m + 1)(m + 2) / 3$.

Inductive step (show P(m + 1))

$\sum (k = 1 \text{ to } m+1)\ k(k + 1) = \sum (k = 1 \text{ to } m)\ k(k + 1) + (m + 1)(m + 2)$

$= m(m + 1)(m + 2) / 3 + (m + 1)(m + 2)$ [by the hypothesis]

$= (m + 1)(m + 2)\ [\ m/3 + 1\ ]$

$= (m + 1)(m + 2)(m + 3) / 3$

This is exactly P(m + 1), since $(m + 1)((m + 1) + 1)((m + 1) + 2) / 3 = (m + 1)(m + 2)(m + 3) / 3$.

Conclusion

P(1) is true and P(m) ⇒ P(m + 1) for every positive integer m. By the principle of mathematical induction, P(n) is true for all positive integers n. ■

4.The given NFA (states q_0, q_1, q_2, q_3 over Σ = {a, b}; q_0 is the start state and q_2 is the only final state) accepts strings that start and end with the same symbol. Construct an equivalent DFA.

Transition table of the given NFA

δ	a	b
→ q_0	$\{q_1\}$	$\{q_3\}$
q_1	$\{q_1, q_2\}$	$\{q_1\}$
* q_2	∅	∅
q_3	$\{q_3\}$	$\{q_3, q_2\}$

Subset construction (start from $\{q_0\}$)

DFA state	on a	on b
→ $\{q_0\}$	$\{q_1\}$	$\{q_3\}$
$\{q_1\}$	$\{q_1, q_2\}$	$\{q_1\}$
$\{q_3\}$	$\{q_3\}$	$\{q_3, q_2\}$
* $\{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_1\}$
* $\{q_3, q_2\}$	$\{q_3\}$	$\{q_3, q_2\}$

Computation: $\delta(\{q_1,q_2\}, a) = \delta(q_1,a) \cup \delta(q_2,a) = \{q_1,q_2\} \cup \emptyset = \{q_1,q_2\}$; $\delta(\{q_1,q_2\}, b) = \{q_1\} \cup \emptyset = \{q_1\}$; $\delta(\{q_3,q_2\}, a) = \{q_3\}$; $\delta(\{q_3,q_2\}, b) = \{q_3,q_2\}$. No new states appear, so the construction stops with 5 states. Every subset containing q_2 is a final state.

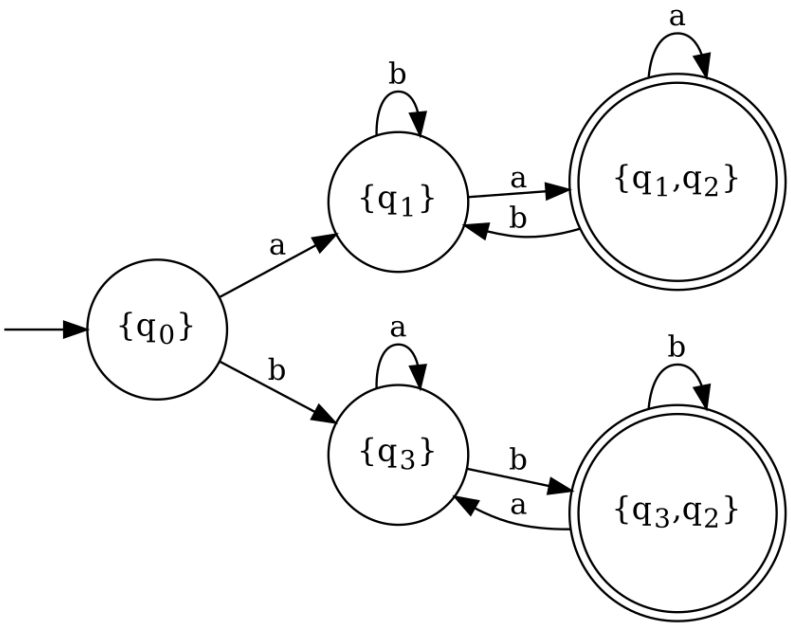
Renaming

$S_0 = \{q_0\}$, $S_1 = \{q_1\}$, $S_2 = \{q_3\}$, $S_3 = \{q_1, q_2\}$, $S_4 = \{q_3, q_2\}$.

δ	a	b
→ S_0	S_1	S_2
S_1	S_3	S_1
S_2	S_2	S_4
* S_3	S_3	S_1
* S_4	S_2	S_4

DFA: $M = (Q, \Sigma, \delta, S_0, F)$ with $Q = \{S_0, S_1, S_2, S_3, S_4\}$, $\Sigma = \{a, b\}$, $F = \{S_3, S_4\}$.

DFA diagram



5. For the given NFA with ε-transitions, find the ε-closure of each state, construct an equivalent NFA without ε-transitions, draw its transition table and mark the initial and final states.

Given ε-NFA (read from the figure)

States {A, B, C}, Σ = {0, 1}, start state A, final state C. Transitions: A on 1 → A; A on ε → B; A on 0 → B; A on 0 → C; B on 1 → B; B on ε → C; C on 0,1 → C.

δ	0	1	ε
→ A	{B, C}	{A}	{B}
B	∅	{B}	{C}
* C	{C}	{C}	∅

ε-closures

- ε-closure(A) = {A, B, C} (A –ε→ B –ε→ C)
- ε-closure(B) = {B, C}
- ε-closure(C) = {C}

Computing δ′(q, a) = ε-closure(δ(ε-closure(q), a))

- δ′(A,0) = ε-closure(δ(A,0) ∪ δ(B,0) ∪ δ(C,0)) = ε-closure({B,C} ∪ ∅ ∪ {C}) = {B, C}
- δ′(A,1) = ε-closure(δ(A,1) ∪ δ(B,1) ∪ δ(C,1)) = ε-closure({A} ∪ {B} ∪ {C}) = {A, B, C}
- δ′(B,0) = ε-closure(δ(B,0) ∪ δ(C,0)) = ε-closure(∅ ∪ {C}) = {C}
- δ′(B,1) = ε-closure(δ(B,1) ∪ δ(C,1)) = ε-closure({B} ∪ {C}) = {B, C}
- δ′(C,0) = ε-closure(δ(C,0)) = {C}
- δ′(C,1) = ε-closure(δ(C,1)) = {C}

Final states

A state becomes final if its ε-closure contains a final state of the original NFA. C is final, and it lies in the ε-closure of A, B and C, so F′ = {A, B, C}. The initial state remains A.

Transition table of the NFA without ε-transitions

δ′	0	1
→* A	{B, C}	{A, B, C}
* B	{C}	{B, C}
* C	{C}	{C}

Transition diagram

